

Questions

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Q1 - 2024 (04 Apr Shift 1)

Let $\alpha, \beta \in \mathbf{R}$. Let the mean and the variance of 6 observations $-3, 4, 7, -6, \alpha, \beta$ be 2 and 23, respectively.

The mean deviation about the mean of these 6 observations is :

(1) $\frac{13}{3}$

(2) $\frac{16}{3}$

(3) $\frac{11}{3}$

(4) $\frac{14}{3}$

Q2 - 2024 (04 Apr Shift 2)

If the mean of the following probability distribution of a random variable X :

X	0	2	4	6	8
P(X)	a	2a	a + b	2b	3b

is $\frac{46}{9}$, then the variance of the distribution is

(1) $\frac{173}{27}$

(2) $\frac{566}{81}$

(3) $\frac{151}{27}$

(4) $\frac{581}{81}$

Q3 - 2024 (05 Apr Shift 1)

From a lot of 10 items, which include 3 defective items, a sample of 5 items is drawn at random. Let the random variable X denote the number of defective items in the sample. If the variance of X is σ^2 , then $96\sigma^2$ is equal to _____

Q4 - 2024 (06 Apr Shift 1)

The mean and standard deviation of 20 observations are found to be 10 and 2, respectively. On rechecking, it was found that an observation by mistake was taken 8 instead of 12. The correct standard deviation is

(1) 1.8

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- (2) 1.94
- (3) $\sqrt{3.96}$
- (4) $\sqrt{3.86}$

Q5 - 2024 (08 Apr Shift 2)

Let $a, b, c \in \mathbb{N}$ and $a < b < c$. Let the mean, the mean deviation about the mean and the variance of the 5 observations 9, 25, a, b, c be 18, 4 and $\frac{136}{5}$, respectively. Then $2a + b - c$ is equal to _____

Q6 - 2024 (09 Apr Shift 1)

The frequency distribution of the age of students in a class of 40 students is given below.

Age	15	16	17	18	19	20
No of Students	5	8	5	12	x	y

If the mean deviation about the median is 1.25, then $4x + 5y$ is equal to :

- (1) 46
- (2) 43
- (3) 44
- (4) 47

Q7 - 2024 (09 Apr Shift 2)

If the variance of the frequency distribution

x	c	$2c$	$3c$	$4c$	$5c$	$6c$
f	2	1	1	1	1	1

is 160, then the value of $c \in \mathbb{N}$ is

- (1) 7
- (2) 8
- (3) 5
- (4) 6

Questions

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Answer Key

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Q1 (1) mathongo Q2 (2) mathongo Q3 (56) mathongo Q4 (3) mathongo

Q5 (33) mathongo Q6 (3) mathongo Q7 (1) mathongo mathongo mathongo

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Solutions

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Q1

$$\frac{\sum x_i}{6} = 2 \text{ and } \frac{\sum x_i^2}{N} - \mu^2 = 23$$

$$\alpha + \beta = 10$$

$$\alpha^2 + \beta^2 = 52$$

solving we get $\alpha = 4, \beta = 6$

$$\frac{\sum |x_i - \bar{x}|}{6} = \frac{5+2+5+8+2+4}{6} = \frac{13}{3}$$

Q2

$$\sum P_i = 1$$

$$a + 2a + a + b + 2b + 3b = 1$$

$$4a + 6b = 1 \dots (I)$$

$$E(x) = \text{mean} = \frac{46}{9}$$

$$\sum P_i X_i = \frac{46}{9} \Rightarrow 4a + 4a + 4b + 12b + 24b = \frac{46}{9}$$

$$8a + 40b = \frac{46}{9}$$

$$4a + 20b = \frac{23}{9} \dots (II)$$

Subtract (I) from (II) we get

$$b = \frac{1}{9} \text{ \& } a = \frac{1}{12}$$

$$\text{Variance} = E(x_i^2) - E(x_i)^2$$

$$E(x_i)^2 = 0^2 \times 9^2 + 2^2 \times 2a + 4^2(a+b) + 6^2(2b) + 8^2(3b) \\ = 24a + 280b$$

$$\text{Put } a = \frac{1}{12} \quad b = \frac{1}{9}$$

$$E(x_i^2) = 2 + \frac{280}{9} = \frac{298}{9}$$

$$\therefore \sigma^2 = E(x_i^2) - E(x_i)^2$$

$$= \frac{298}{9} - \left(\frac{46}{9}\right)^2$$

$$\sigma^2 = \frac{298}{9} - \frac{2116}{81}$$

$$= \frac{566}{81}$$

Q3

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Solutions

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 X = denotes number of defective

x	0	1	2	3
P(x)	$\frac{7}{15}$	$\frac{5}{12}$	$\frac{5}{12}$	$\frac{1}{12}$
x_1^2	0	1	4	9
$P_i x_1^2$	0	$\frac{5}{12}$	$\frac{20}{12}$	$\frac{9}{12}$
$p_i x_i$	0	$\frac{5}{12}$	$\frac{10}{12}$	$\frac{3}{12}$

$$\mu = \sum p_i x_i = \frac{18}{12}$$

$$\sum p_i x_1^2 = \frac{34}{12}$$

$$\sigma^2 = \sum p_i x_1^2 - (\mu)^2$$

$$= \frac{34}{12} - \left(\frac{18}{12}\right)^2 = \frac{17}{6} - \frac{9}{4}$$

$$\frac{34 - 27}{12} = \frac{7}{12}$$

$$96\sigma^2 = 96 \times \frac{7}{12} = 56$$

Q4

Mean ($\bar{x}$) = 10

$$\Rightarrow \frac{\sum x_i}{20} = 10$$

$$\sum x_i = 10 \times 20 = 200$$

If 8 is replaced by 12, then $\sum x_i = 200 - 8 + 12 = 204$

$$\therefore \text{Correct mean } (\bar{x}) = \frac{\sum x_i}{20}$$

$$= \frac{204}{20} = 10.2$$

$$\therefore \text{Standard deviation} = 2$$

$$\therefore \text{Variance} = (\text{S.D.})^2 = 2^2 = 4$$

$$\Rightarrow \frac{\sum x_i^2}{20} - \left(\frac{\sum x_i}{20}\right)^2 = 4$$

$$\Rightarrow \frac{\sum x_i^2}{20} - (10)^2 = 4$$

$$\Rightarrow \frac{\sum x_i^2}{20} = 104$$

$$\Rightarrow \sum x_i^2 = 2080$$

Now, replaced '8' observations by '12'

$$\text{Then, } \sum x_i^2 = 2080 - 8^2 + 12^2 = 2160$$

 $\therefore$ Variance of removing observations

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Solutions

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$$\Rightarrow \frac{\sum x_i^2}{20} - \left(\frac{\sum x_i}{20} \right)^2$$

$$\Rightarrow \frac{2160}{20} - (10.2)^2$$

$$\Rightarrow 108 - 104.04$$

$$\Rightarrow 3.96$$

Correct standard deviation

$$= \sqrt{3.96}$$

Q5

$$a, b, c \in N \quad a < b < c$$

$$\bar{x} = \text{mean} = \frac{9+25+a+b+c}{5} = 18$$

$$a + b + c = 56$$

$$\text{Mean deviation} = \frac{\sum |x_i - \bar{x}|}{n} = 4$$

$$= 9 + 7 + |18 - a| + |18 - b| + |18 - c| = 20$$

$$= |18 - a| + |18 - b| + |18 - c| = 4$$

$$\text{Variance} = \frac{\sum |x_i - \bar{x}|^2}{n} = \frac{136}{5}$$

$$= 81 + 49 + |18 - a|^2 + |18 - b|^2 + |18 - c|^2 = 136$$

$$= (18 - a)^2 + (18 - b)^2 + (18 - c)^2 = 6$$

$$\text{Possible values } (18 - a)^2 = 1, \quad (18 - b)^2 = 1 \quad (18 - c)^2 = 4 \quad a < b < c$$

$$\text{so } 18 - a = 1 \quad 18 - b = -1 \quad 18 - c = -2$$

$$a = 17 \quad b = 19 \quad c = 20$$

$$a + b + c = 56$$

$$2a + b - c = 19 - 20 = -1$$

Q6

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Solutions

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$$x + y = 10 \dots\dots\dots (1)$$

$$\text{Median} = 18 = M$$

$$\text{M.D.} = \frac{\sum f_i |x_i - M|}{\sum f_i}$$

$$1.25 = \frac{36 + x + 2y}{40}$$

$$x + 2y = 14 \dots\dots\dots (2)$$

by (1) & (2)

$$x = 6, y = 4$$

$$\Rightarrow 4x + 5y = 24 + 20 = 44$$

Age(x_i)	f	$ x_i - M $	$f_i x_i - M $
15	5	3	15
16	8	2	16
17	5	1	5
18	12	0	0
19	x	1	x
20	y	2	2y

Q7

x	C	2C	3C	4C	5C	6C
f	2	1	1	1	1	1

$$\bar{x} = \frac{(2+2+3+4+5+6)C}{7} = \frac{22C}{7}$$

$$\text{Var}(x) = \frac{c^2 (2^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2)}{7}$$

$$- \left(\frac{22c}{7} \right)^2$$

$$= \frac{92c^2}{7} - c^2 \times \frac{484}{49}$$

$$= \frac{(644 - 484)c^2}{49} = \frac{160c^2}{49}$$

$$160 = \frac{160 \times c^2}{49} \Rightarrow c = 7$$

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